

Dataset Validation Framework Report

Target Dataset: **synthetic_data/v2.2_production_gold** | Verdict: **PASS**

Metric	Category	Measured Value	Threshold	Status
GM Positive Ratio	Physics	420/420	100%	PASS
NaN / Inf Free Files	Physics	420/420	100%	PASS
Speed Sanity Ratio	Physics	420/420	100%	PASS
Scenario Roll Response Compliance	Scenario	388/420	> 85%	PASS
Resonance Frequency Alignment	Scenario	371/420	> 80%	PASS
Dynamic Channel Variance Check	Statistical	Passed (0 frozen)	0 frozen channels	PASS
Extreme Outliers (< 120° Roll)	Statistical	420/420	100%	PASS
Temporal Continuity Compliance	Time-Series	420/420	100%	PASS
Sensor Flatline Check	Time-Series	420/420	100%	PASS
Class Balance: p_sync	Label	108 files > 0.50	>= 15 files	PASS
Class Balance: p_param	Label	102 files > 0.50	>= 15 files	PASS
Class Balance: p_broach	Label	66 files > 0.50	>= 15 files	PASS
Class Balance: p_pure_loss	Label	59 files > 0.50	>= 15 files	PASS
Class Balance: p_dead_ship	Label	30 files > 0.50	>= 15 files	PASS
Vessel Class Diversity	Diversity	30 classes	>= 3 classes	PASS
Sea State Height Coverage	Diversity	1.3m - 10.2m	Hs max >= 6.0m	PASS
Sensor Noise / Packet Loss Stress Test	Sensor	PASSED	Pass stress simulation	PASS

Failures and Recommendations

No failures detected.